

Chapter 2

Table 2-1: Parallel Port Signals, arranged by pin number.

Pin: D-sub	Signal	Function	Source	Register		Inverted at con- nector?	Pin: Centron- ics
				Name	Bit #		
1	nStrobe	Strobe D0-D7	PC ¹	Control	0	Y	1
2	D0	Data Bit 0	PC ²	Data	0	N	2
3	D1	Data Bit 1	PC ²	Data	1	N	3
4	D2	Data Bit 2	PC ²	Data	2	N	4
5	D3	Data Bit 3	PC ²	Data	3	N	5
6	D4	Data Bit 4	PC ²	Data	4	N	6
7	D5	Data Bit 5	PC ²	Data	5	N	7
8	D6	Data Bit 6	PC ²	Data	6	N	8
9	D7	Data Bit 7	PC ²	Data	7	N	9
10	nAck	Acknowledge (may trigger interrupt)	Printer	Status	6	N	10
11	Busy	Printer busy	Printer	Status	7	Y	11
12	PaperEnd	Paper end, empty (out of paper)	Printer	Status	5	N	12
13	Select	Printer selected (on line)	Printer	Status	4	N	13
14	nAutoLF	Generate automatic line feeds after carriage returns	PC ¹	Control	1	Y	14
15	nError (nFault)	Error	Printer	Status	3	N	32
16	nInit	Initialize printer (Reset)	PC ¹	Control	2	N	31
17	nSelectIn	Select printer (Place on line)	PC ¹	Control	3	Y	36
18	Gnd	Ground return for nStrobe, D0					19,20
19	Gnd	Ground return for D1, D2					21,22
20	Gnd	Ground return for D3, D4					23,24
21	Gnd	Ground return for D5, D6					25,26
22	Gnd	Ground return for D7, nAck					27,28
23	Gnd	Ground return for nSelectIn					33
24	Gnd	Ground return for Busy					29
25	Gnd	Ground return for nInit					30
	Chassis	Chassis ground					17
	NC	No connection					15,18,34
	NC	Signal ground					16
	NC	+5V	Printer				35

¹Setting this bit high allows it to be used as an input (SPP only).

²Some Data ports are bidirectional.

The signal names in the tables are those used by the parallel port in the original IBM PC. The names describe the signals' functions in PC-to-peripheral transfers. In other modes, the functions and names of many of the signals change.

Table 2-3: Differences between original Centronics interface and PC interface

Pin (Centronics)	Original Function	New (PC) Function
14	signal ground	nAutoLF
15	oscillator out	no connection
16	signal ground	no connection
17	chassis ground	no connection
18	+5V	no connection
33	light detect	Ground return for nSelectIn
34	line count	no connection
35	Ground return for line count	no connection
36	Reserved	nSelectIn

The PC's D-sub connector has just 25 contacts, compared to the Centronics connector's 36. Six of the original Centronics signals have no connection at the PC, and the PC has five fewer ground-return pins.

The PC interface also redefines three signals. Pin 14 (*Signal Ground*) is nAutoLF on the PC, pin 36 (*Reserved*) is nSelectIn, and pin 33 (*Light Detect*) is the ground return for nSelectIn.

that the register bit is the inverse of the signal at the connector, but the name says nothing about the signal's purpose. Whether to use *nStrobe* or *C0* depends on which type of information is more relevant to the topic at hand.

The Data Register

The Data port, or Data register, (*D0-D7*) holds the byte written to the Data outputs. In bidirectional Data ports, when the port is configured as input, the Data register holds the byte read at the connector's Data pins. Although the Centronics interface and the IEEE-1284 standard refer to the Data lines as *D1* through *D8*, in this book, I use *D0-D7* throughout, to correspond to the register bits.

The Status Register

The Status port, or Status register, holds the logic states of five inputs, *S3* through *S7*. Bits *S0-S2* don't appear at the connector. The Status register is read-only, except for *S0*, which is a timeout flag on ports that support EPP transfers, and can be cleared by software. On many ports, the Status inputs have pull-up resistors. In their conventional uses, the Status bits have the following functions:

S0: *Timeout*. In EPP mode, this bit may go high to indicate a timeout of an EPP data transfer. Otherwise unused. This bit doesn't appear on the connector.

S1: Unused.